

Human-AI Collaboration in Scientific Writing and Research Workflows: A Survey

Large language models and other AI systems are changing scientific work by supporting ideation, literature search, reading, writing, coding, data analysis, experiment planning, peer review, and research management. These tools create opportunities for productivity and access, but they also introduce risks around accuracy, authorship, hallucination, attribution, overreliance, bias, and research integrity. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around AI-assisted scientific writing, literature review and citation support, peer review, research agents, human-AI interaction design, AI-for-science workflows, ethics, and evaluation. The survey argues that productive human-AI research collaboration requires workflow-aware systems, transparent provenance, calibrated trust, and domain-specific evaluation.

Keywords:

human-AI collaboration; scientific writing; research workflows; literature review; peer review; research agents; AI for science; research integrity

1 Introduction

Scientific work is a sequence of interdependent activities: formulating questions, reading literature, designing studies, collecting and analyzing data, writing manuscripts, reviewing claims, responding to feedback, and maintaining provenance. AI systems now touch each of these activities. Large language models can draft text, summarize papers, generate code, suggest hypotheses, inspect tables, help with peer review, and coordinate multi-step research workflows [1, 9, 70, 115, 167, 219, 237, 297].

Human-AI collaboration in science is not simply automation. Researchers need systems that preserve expertise, uncertainty, accountability, and creativity. A useful assistant must help with routine work without fabricating evidence, erasing authorship, hiding provenance, or encouraging superficial scholarship. The research workflow is therefore a socio-technical object: models, tools, interfaces, norms, institutions, and evaluation protocols all shape whether collaboration improves science [1, 70, 104, 152, 237, 265, 282, 297].

This survey reviews the literature on scientific writing and research workflows from a human-AI collaboration perspective.

We connect writing tools, retrieval systems, research agents, peer-review support, AI-for-science workflows, and governance.

2 Background

2.1 Scientific Writing as Knowledge Work

Scientific writing is not only text production. It involves selecting claims, linking evidence, positioning contributions, choosing terminology, citing prior work, and responding to community expectations. AI writing assistants can help with drafting, revision, structure, grammar, and summarization, but they can also introduce unsupported claims or homogenize scientific voice. Effective systems must therefore expose sources, uncertainty, and editable rationales [100, 115, 149, 214, 219, 237, 255, 297].

2.2 Literature Review and Citation Workflows

Literature review requires search, screening, reading, synthesis, citation management, and gap iden-

tification. Retrieval-augmented and LLM-based tools can summarize papers and recommend citations, but they must handle incomplete coverage, citation bias, stale corpora, and errors in attribution. Systematic reviews require especially careful audit trails and reproducible inclusion decisions [1, 9, 115, 149, 162, 167, 214, 219].

2.3 Peer Review and Scholarly Evaluation

AI can assist reviewers by summarizing manuscripts, checking claims, identifying missing related work, and drafting feedback. It can also help authors anticipate reviewer questions. However, peer review depends on confidentiality, expert judgment, accountability, and fairness. AI-generated reviews risk shallow criticism, leakage of confidential manuscripts, and hidden bias unless governance is explicit [9, 70, 115, 214, 219, 237, 255, 267].

2.4 Research Agents and Workflow Automation

Research agents combine models with tools such as search engines, code execution, data analysis libraries, theorem provers, laboratory APIs, and document editors. They can coordinate multi-step workflows, but they also compound errors: a flawed search can produce weak evidence, which can lead to a misleading draft or experiment plan. Human oversight and workflow checkpoints are therefore central [1, 70, 100, 115, 149, 219, 237, 297].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with human-AI collaboration, scientific writing, academic writing, writing assistants, LLMs, ChatGPT, manuscript drafting, scientific document generation, literature review, systematic review, scholarly search, citation generation, scientific summarization, research assistants, research agents, AI scientists, autonomous research, scientific workflows, AI for science, scientific discovery, hypothesis generation, experiment design, peer review, automated reviewing, AI authorship, research integrity, LLM-generated scientific text, co-creation, mixed-initiative interaction, and user studies. Records were retained when ti-

tle or abstract evidence connected AI or human-AI systems to scientific writing, literature work, peer review, research automation, AI-for-science workflows, authorship, integrity, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *workflow stage*: ideation, search, reading, synthesis, data analysis, experiment design, writing, review, publication, or maintenance. The second is *AI role*: autocomplete, editor, retriever, summarizer, critic, planner, agent, collaborator, or auditor. The third is *human role*: author, reviewer, supervisor, domain expert, data steward, or responsible approver. The fourth is *artifact*: query, note, citation, figure, table, code, dataset, manuscript, review, rebuttal, or provenance log. The fifth is *risk*: hallucination, citation error, bias, plagiarism, confidentiality breach, authorship ambiguity, or skill erosion. The sixth is *interface*: conversational, document-integrated, notebook-based, agentic, or mixed-initiative. The seventh is *evaluation*: productivity, quality, factuality, reproducibility, user trust, learning, and downstream scientific validity. The eighth is *governance*: disclosure, consent, auditability, policy, and accountability.

3.3 Writing and Revision Support

Writing support ranges from local edits to full manuscript drafting. The central design issue is control: researchers need assistance that accelerates expression while preserving intellectual responsibility. Good tools should cite evidence, distinguish generated from human text, support versioning, and make it easy to reject or revise suggestions [100, 115, 149, 214, 219, 237, 255, 297].

3.4 Retrieval, Reading, and Synthesis

Literature tools connect search, retrieval, summarization, and citation. They should help users inspect evidence rather than simply produce plausible summaries. Source-grounded synthesis, transparent inclusion criteria, and citation-level verification are especially important when tools support systematic reviews or manuscript introductions [1, 9, 115, 149, 162, 167, 214, 219].

3.5 Peer Review Assistance

Peer-review tools can check clarity, novelty, related work, method consistency, and reproducibility artifacts. They are most useful as scaffolds for human judgment rather than replacements for expertise. Evaluation should ask whether AI improves review quality, fairness, and timeliness without compromising confidentiality or accountability [9, 70, 115, 214, 219, 237, 255, 267].

3.6 Research Agents

Agentic systems can plan literature searches, run code, analyze data, generate figures, and draft reports. Their value depends on tool reliability, memory, provenance, and human approval gates. Research agents should record intermediate states so that users can audit which sources, code, datasets, and assumptions produced a claim [1, 70, 100, 115, 149, 219, 237, 297].

3.7 Interaction Design

Human-AI research systems need interaction patterns that fit scientific work: editable suggestions, side-by-side source inspection, uncertainty displays, provenance trails, task-specific controls, and interruption-aware workflows. User studies should examine how systems affect reasoning, not just completion time [1, 70, 104, 152, 237, 265, 282, 297].

3.8 AI for Science Workflows

AI-for-science systems extend collaboration beyond writing into hypothesis generation, experiment design, simulation, laboratory planning, and data analysis. These workflows require domain constraints, validation loops, and integration with instruments or scientific software. Human expertise remains central because scientific plausibility and feasibility cannot be inferred from text alone [4, 144, 158, 161, 188, 219, 236, 297].

3.9 Ethics and Governance

Scientific AI use raises questions about authorship, disclosure, plagiarism, confidentiality, bias, and responsibility. Policies should distinguish assistance, generation, analysis, and autonomous decision-making. Governance should focus not only on whether AI was used, but on whether claims are traceable, sources are verified, and humans remain accountable [4, 9, 70, 85, 162, 196, 237, 267].

3.10 Evaluation Methodology

Evaluation should combine task metrics with scientific validity. Useful measures include factual accuracy, citation correctness, review helpfulness, reproducibility, provenance completeness, user calibration, productivity, and downstream quality. Benchmarks should include realistic workflows rather than isolated prompts, because many failures arise across multi-step chains [9, 70, 115, 149, 167, 219, 237, 297].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across writing, literature review, peer review, research agents, interaction design, AI-for-science workflows, ethics, and evaluation.

Ai-assisted scientific writing and editing The AI-assisted scientific writing and editing cluster contains work that develops methods, systems, datasets, or analyses central to this survey [49, 92, 97, 100, 115, 129, 177, 274, 296]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [33, 43, 134, 147, 155, 187, 189, 206, 243]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 54, 60, 68, 82, 124, 128, 136, 204]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [31, 48, 58, 90, 151, 160, 169, 247, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [65, 98, 114, 130, 146, 148, 226, 266, 300].

Literature review, retrieval, and citation support The literature review, retrieval, and citation support cluster contains work that develops methods, systems, datasets, or analyses central to this survey [1, 2, 32, 63, 178, 214, 255, 267, 285]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [21, 76, 96, 113, 199, 238, 241, 262, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 5, 17, 111, 140, 154, 228, 235, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 19, 45, 91, 122, 145, 205, 290, 293]. Additional retained studies broaden the cluster with alterna-

tive architectures, training signals, data sources, and deployment assumptions [27, 50, 64, 173, 251, 269, 280, 286, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 75, 126, 164, 220, 224, 248, 283, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 73, 78, 180, 202, 209, 210, 244, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 39, 42, 105, 138, 182, 215, 278, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 13, 57, 66, 116, 165, 198, 232, 246]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [20, 69, 81, 108, 183–185, 191, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [71, 80, 83, 125, 166, 194, 223, 229, 231]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 46, 47, 61, 103, 112, 170, 216, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [35, 53, 67, 222].

Peer review and scholarly evaluation The peer review and scholarly evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [119, 161, 292].

Research agents and workflow automation The research agents and workflow automation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [9, 70, 144, 149, 188, 203, 219, 237, 297]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [85, 104, 153, 158, 181, 211, 236, 265, 277]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [59, 123, 156, 176, 239, 258, 261, 264, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 44, 74, 89, 93, 250, 256, 275, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [137, 240, 245].

Human-ai collaboration and interaction design The human-AI collaboration and interaction design cluster contains work that develops methods, systems, datasets, or analyses central to this survey [152, 242].

AI for science and discovery workflows The AI for science and discovery workflows cluster contains work that develops methods, systems, datasets, or analyses central to this survey [40, 110, 249, 263].

Ethics, authorship, integrity, and governance The ethics, authorship, integrity, and governance cluster contains work that develops methods, systems, datasets, or analyses central to this survey [4, 10, 11, 25, 28, 62, 131, 135, 172]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [29, 37, 56, 72, 174, 200, 208, 227, 233]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [41, 95, 99, 150].

Benchmarks, evaluation, and productivity The benchmarks, evaluation, and productivity cluster contains work that develops methods, systems, datasets, or analyses central to this survey [55, 87, 157, 162, 167, 179, 196, 218, 282]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [16, 79, 106, 190, 217, 234, 253, 254, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [36, 77, 94, 102, 117, 141, 159, 197, 260]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [120, 132, 142, 175, 212, 225, 252, 268, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 51, 52, 84, 88, 107, 127, 221, 230]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 86, 121, 139, 163, 171, 213, 271, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [8, 24, 34, 109, 143, 186, 192, 195, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [101, 118, 133, 168, 193, 201, 207, 272, 276]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [38].

4 Challenges and Prospects

4.1 Grounded Assistance Rather Than Plausible Text

Scientific writing tools must prioritize evidence grounding, citation correctness, and traceability. Future systems should make unsupported claims difficult to introduce and easy to detect.

4.2 Workflow-Level Evaluation

Most evaluations isolate a prompt or output. Scientific work is multi-step, so benchmarks should test how AI assistance changes literature coverage, experimental choices, analysis errors, writing quality, and review decisions over time.

4.3 Human Expertise and Skill Preservation

AI systems should augment scientific judgment rather than replace it. Interfaces should encourage inspection, critique, and learning, especially for early-career researchers.

4.4 Governance and Disclosure

Institutions need policies for authorship, confidentiality, peer review, data handling, and provenance. Disclosure should be specific enough to make scientific artifacts auditable without reducing policy to a simple yes-or-no AI-use label.

4.5 Research Agents with Auditable Provenance

Agentic research systems will become more useful as they integrate tools, memory, and scientific software. Their success will depend on transparent logs, approval gates, validation tests, and source-aware generation.

5 Conclusion

Human-AI collaboration in scientific writing and research workflows is reshaping how researchers search, read, write, review, analyze, and discover. This survey reviewed the literature across writing assistance, literature synthesis, peer review, research agents, interaction design, AI-for-science workflows, ethics, and evaluation. The main lesson is that scientific collaboration with AI must be workflow-aware and evidence-grounded: productivity gains matter only

when claims remain traceable, humans remain accountable, and scientific quality improves.

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